The Negative Binomial Distribution

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Introduction

The negative binomial distribution is the workhorse model for overdispersed counts – counts whose variance exceeds their mean. It arises naturally as the number of failures before the r-th success in a sequence of Bernoulli trials, or as a gamma mixture of Poisson distributions. It dominates modern count-regression practice (RNA-seq, public-health incidence, insurance claims).

Prerequisites

Geometric and Poisson distributions; gamma distribution is useful for the mixture interpretation.

Theory

Two standard parameterisations:

Counts-of-failures form: Y is the number of failures before the r-th success in iid Bernoulli(p) trials:

$$P(Y = k) = \binom{k + r - 1}{k} p^r (1 - p)^k, \qquad k = 0, 1, 2, \ldots$$

Mean-dispersion form: parameterise by mean μ and dispersion (size) parameter θ with

$$P(Y = k) = \frac{\Gamma(k + \theta)}{k!\,\Gamma(\theta)} \left(\frac{\theta}{\theta + \mu}\right)^{\theta} \left(\frac{\mu}{\theta + \mu}\right)^{k}.$$

Moments in the mean-dispersion form:

- $E[Y] = \mu$.
- $\mathrm{Var}(Y) = \mu + \mu^2/\theta$.

Gamma-Poisson interpretation: if $\Lambda \sim \mathrm{Gamma}(\theta, \theta/\mu)$ and $Y \mid \Lambda \sim \mathrm{Poisson}(\Lambda)$, then marginally $Y \sim \mathrm{NegBin}(\mu, \theta)$. As $\theta \to \infty$, the NB collapses to Poisson(μ).

Geometric as a special case: $r = 1$ (or $\theta = 1$) recovers the geometric distribution.

Assumptions

No common-rate assumption (unlike Poisson); the heterogeneity in underlying rates is parameterised by θ.

R Implementation

```r
library(MASS)

# R's rnbinom: parameterise by size (= theta) and mu
mu <- 5; theta <- 2
Y <- rnbinom(1e5, size = theta, mu = mu)
```

```r
c(mean     = mean(Y),
  variance = var(Y),
  expected_var = mu + mu^2 / theta)

# PMF comparison to Poisson
k <- 0:20
pmf_nb   <- dnbinom(k, size = theta, mu = mu)
pmf_pois <- dpois(k, lambda = mu)

data.frame(k, nb = round(pmf_nb, 4), poisson = round(pmf_pois, 4))

# Fit by MLE using MASS::glm.nb
df <- data.frame(y = Y, x = rnorm(1e5))
fit <- glm.nb(y ~ x, data = df)
summary(fit)$theta
```

Output & Results

```
     mean  variance expected_var
    5.020    17.48        17.50

    k      nb  poisson
1   0  0.2041   0.0067
2   1  0.2041   0.0337
3   2  0.1701   0.0842
4   3  0.1296   0.1404
5   4  0.0925   0.1755
6   5  0.0634   0.1755
...

[1] 1.998        # estimated theta
```

Empirical variance (17.5) matches the theoretical $\mu + \mu^2/\theta = 5 + 25/2 = 17.5$. The MLE of θ is close to the true value 2.

Interpretation

Negative binomial is the default count-data model in modern applied work. In a manuscript: "incidence rates were compared using negative binomial regression to account for overdispersion (dispersion estimated as $\theta = 2.0$)". Always report the dispersion parameter alongside the rate ratios.

Practical Tips

- Detect overdispersion by fitting a Poisson first and checking residual deviance $\approx$ df; ratios > 1.5 suggest overdispersion.
- `glm.nb` jointly estimates regression coefficients and dispersion; faster-but-less-flexible alternatives include `glmmTMB` and `brms`.
- Zero-inflated variants (`pscl::zeroinfl` with `dist = "negbin"`) handle excess zeros on top of overdispersion.
- Reparameterisations: some textbooks use (n, p), some (μ, θ); R's `rnbinom(n, size, prob)` or `rnbinom(n, size, mu)` accepts either.
- In RNA-seq, DESeq2 and edgeR both model counts as negative binomial with shrinkage on the dispersion parameter.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Probability, conditional probability, Bayes' theorem
- Discrete distributions: Bernoulli, binomial, Poisson
- Continuous distributions and Q-Q plots